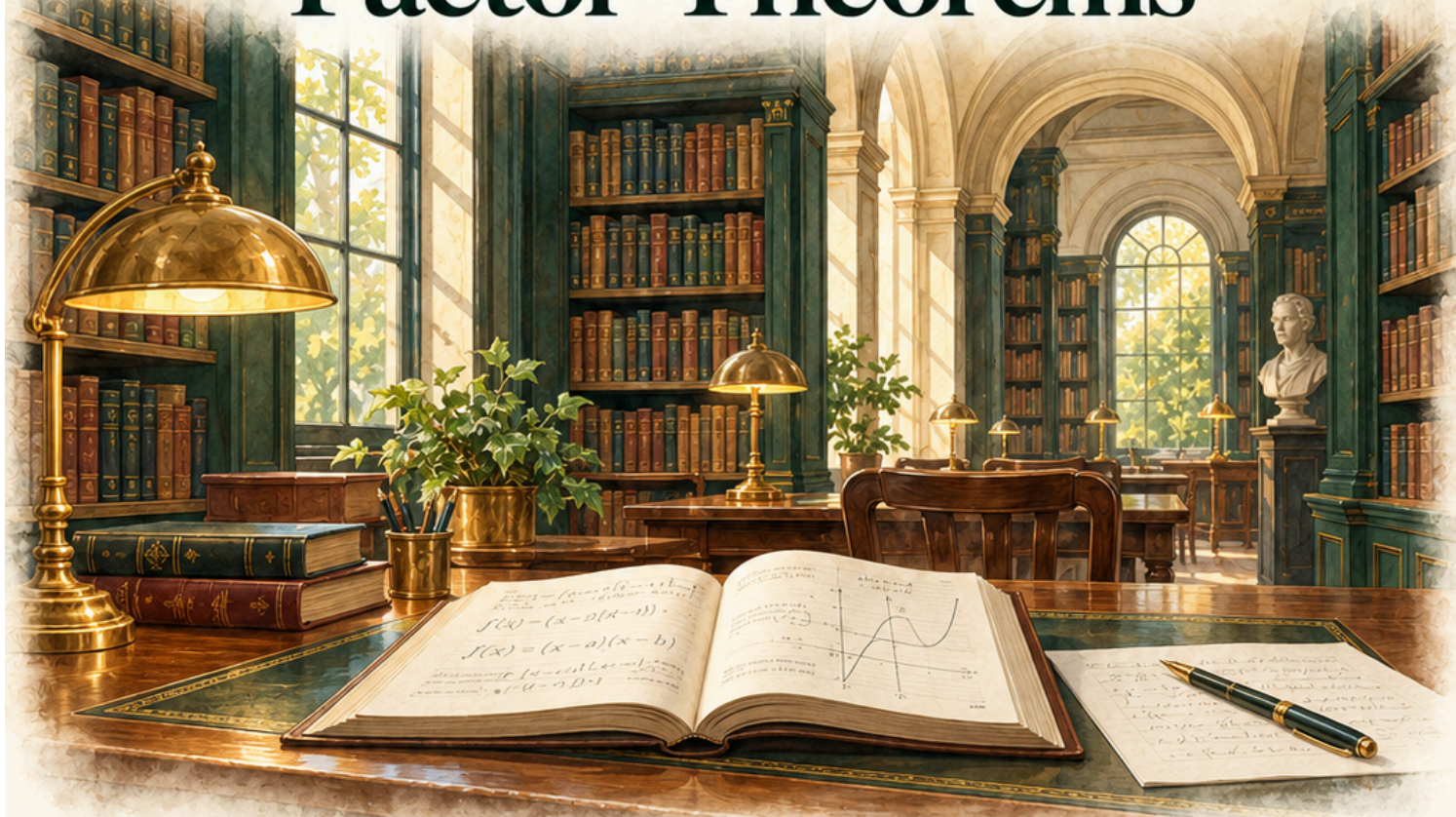




ALGEBRA II
CLASS 11

Factoring Polynomials & the Remainder and Factor Theorems



Recognize
patterns



Factor
completely



Connect factors
and remainders

Build understanding. Discover connections.

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Item 1 – Factoring Polynomials

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What is factoring?

Factoring means rewriting a polynomial as a product of simpler expressions.



In simple terms

We break a polynomial into factors that multiply together to make the original expression.

Example 1

Factor the polynomial.

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

2 and 3 multiply to 6 and add to 5.

Example 2

Factor out the greatest common factor (GCF).

$$6x^3 + 9x^2 = 3x^2(2x + 3)$$

Both terms share a GCF of $3x^2$.



Why this matters

Factoring helps us simplify expressions, solve equations, and find zeros.



Quick check

Multiply the factors back together to make sure you get the original polynomial.

Image 1 practice: What factoring polynomials means

Q1. Factor completely over the integers: $5x + 35$.

Q2. Factor completely over the integers: $x^2 + 11x + 28$.

Q3. Does $(x + 2)(x + 7)$ equal $x^2 + 9x + 12$? Expand and, if needed, correct the constant term.

Q4. Factor completely over the integers: $3x^2 - 27$.

Q5. Factor completely over the integers: $2x^3 + 14x^2 + 24x$.

Item 1A – Greatest Common Factor (GCF)

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What is GCF?

The greatest common factor is the largest factor shared by all the terms of a polynomial.



In simple terms

Find the biggest number and variable part that every term has in common, then factor it out.

Example 1

Factor out the greatest common factor.

$$6x^3 + 9x^2 = 3x^2(2x + 3)$$

The GCF is $3x^2$.

Example 2

Find the GCF and factor.

$$8x^2y + 12xy^2 = 4xy(2x + 3y)$$

The GCF is $4xy$.



How to find it

- 1 Find the greatest common factor of the coefficients.
- 2 Take each shared variable with the lowest exponent.
- 3 Factor that GCF out of every term.



Quick check

Distribute the factor back to make sure you get the original polynomial.

Image 2 practice: Greatest common factor

Q6. Factor completely over the integers: $14x + 42$.

Q7. Factor completely over the integers: $20x^3 - 30x^2$.

Q8. Factor completely over the integers: $18x^2y + 24xy^2$.

Q9. Factor completely over the integers: $-12x^3 + 20x^2 - 8x$.

Q10. Factor completely over the integers: $15x^4y^2 - 25x^3y^3 + 10x^2y^4$.

Item 1B — Factoring Trinomials

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What is a trinomial?

A trinomial is a polynomial with three terms. Factoring a trinomial means rewriting it as a product of simpler binomials.



In simple terms

We look for numbers that multiply to the last term and add to the middle-term coefficient.

Example 1

Factor the trinomial.

$$x^2 + 7x + 12 = (x + 3)(x + 4)$$

3 and 4 multiply to 12 and add to 7.

Example 2

Factor the trinomial.

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

2 and 3 multiply to 6 and add to 5.



How to do it

- 1 Find two numbers that multiply to c .
- 2 Check that the same two numbers add to b .
- 3 Write the trinomial as two binomials.



Quick check

Multiply the factors back together to make sure you get the original trinomial.

Image 3 practice: Factoring trinomials

Q11. Factor completely over the integers: $x^2 + 9x + 18$.

Q12. Factor completely over the integers: $x^2 - 11x + 24$.

Q13. Factor completely over the integers: $x^2 + 2x - 35$.

Q14. Factor completely over the integers: $2x^2 + 9x + 10$.

Q15. Factor completely over the integers: $6x^2 - x - 12$.

Item 1C – Difference of Squares

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What is a difference of squares?

A difference of squares is binomial with two squared terms separated by subtraction.



In simple terms

If both terms are perfect squares and there is a minus sign, we can factor using a special pattern.

Example 1

Factor the expression.

$$x^2 - 25 = (x - 5)(x + 5)$$

$25 = 5^2$, so this matches $a^2 - b^2$.

Example 2

Factor the expression.

$$9x^2 - 16 = (3x - 4)(3x + 4)$$

$9x^2 = (3x)^2$ and $16 = 4^2$.



Pattern

$$a^2 - b^2 = (a - b)(a + b)$$

Take the square root of each term. Then write one binomial with a minus sign and one with a plus sign.



Quick check

Multiply the factors back together to make sure you get the original expression.

$$(x - 5)(x + 5) = x^2 - 25$$

$$a^2 - b^2$$

Image 4 practice: Difference of squares

Q16. Factor completely over the integers: $x^2 - 49$.

Q17. Factor completely over the integers: $16x^2 - 81$.

Q18. Factor completely over the integers: $25x^2 - 36y^2$.

Q19. Factor completely over the integers: $3x^2 - 147$.

Q20. Factor completely over the integers: $x^4 - 256$.

Item 1D – Perfect-Square Trinomials

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What is a perfect-square trinomial?

A perfect-square trinomial is a trinomial that comes from squaring a binomial.



In simple terms

If the first and last terms are perfect squares and the middle term matches $2ab$, the trinomial can be written as a squared binomial.

Example 1

Factor the trinomial.

$$x^2 + 10x + 25 = (x + 5)^2$$

$$x^2 = x^2, 25 = 5^2, \text{ and } 10x = 2 \cdot x \cdot 5.$$

Example 2

Factor the trinomial.

$$x^2 - 6x + 9 = (x - 3)^2$$

$$x^2 = x^2, 9 = 3^2, \text{ and } -6x = -2 \cdot x \cdot 3.$$



Pattern

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

Look for two perfect squares and a middle term equal to $\pm 2ab$.



Quick check

Expand the binomial to make sure it matches the original trinomial.

$$(x + 5)^2 = x^2 + 10x + 25$$

$$x^2 \pm 2xy + y^2$$

Image 5 practice: Perfect-square trinomials

Q21. Factor completely over the integers: $x^2 + 14x + 49$.

Q22. Factor completely over the integers: $x^2 - 16x + 64$.

Q23. Factor completely over the integers: $9x^2 + 24x + 16$.

Q24. Factor completely over the integers: $12x^2 - 60x + 75$.

Q25. Find both values of k for which $16x^2 + kx + 81$ is a perfect-square trinomial, and factor each version.

Item 1E – Factoring by Grouping

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What is factoring by grouping?

Factoring by grouping is a method used for polynomials with four terms. We group the terms in pairs, factor out the greatest common factor (GCF) from each pair, and then factor out a common binomial factor.



When to use it?

Use factoring by grouping when a polynomial has four terms and there is a common binomial factor after factoring each pair.

It is very common with four terms.

Example 1

Factor the polynomial.

Step 1: Group the terms in pairs.

$$x^3 + 3x^2 + 2x + 6 = (x^3 + 3x^2) + (2x + 6)$$

Step 2: Factor out the GCF from each pair.

$$x^2(x + 3) + 2(x + 3)$$

Step 3: Factor the common binomial.

$$(x + 3)(x^2 + 2)$$

Group the first two terms and the last two terms.

Factor out x^2 from the first pair and 2 from the second pair.

Both terms have the common binomial $(x + 3)$. Factor it out.



Pattern

For a polynomial with four terms:

$$ax^3 + bx^2 + cx + d$$

1. Group the terms in pairs.
2. Factor out the GCF from each pair.
3. Look for a common binomial factor.
4. Factor it out.



Quick check

Factor the polynomial.

$$2x^3 + 4x^2 + 3x + 6$$

Hint: Group, factor each pair, then look for a common binomial.

Image 6 practice: Factoring by grouping

Q26. Factor completely over the integers: $x^3 + 5x^2 + 3x + 15$.

Q27. Factor completely over the integers: $2x^3 + 8x^2 + 5x + 20$.

Q28. Factor completely over the integers: $x^3 - 4x^2 - 9x + 36$.

Q29. Factor completely over the integers: $3x^3 - 6x^2 - 12x + 24$.

Q30. Factor completely over the integers: $2x^3 + 3x^2 - 8x - 12$.

2. Factoring Helps Find Zeros

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Why is this important?

Factoring a polynomial helps us find the **zeros** (also called **roots** or **solutions**) of a function. The zeros are the x -values where the graph crosses the x -axis (i.e., where $f(x) = 0$).



Key Idea

If a polynomial is written in factored form,

$$f(x) = (x - a)(x - b)$$

then the zeros are
 $x = a$ or $x = b$.

Example 1

Find the zeros of the function.

Suppose $f(x) = x^2 - 5x + 6$.

Factor: $f(x) = (x - 2)(x - 3)$.

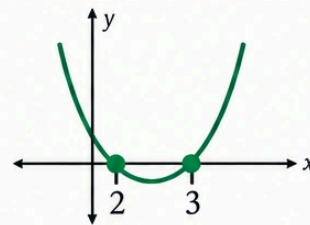
To find the zeros, set the function equal to zero:

$$(x - 2)(x - 3) = 0$$

Therefore: $x = 2$ or $x = 3$.

These are the **roots**, **zeros**, or **solutions**.

Check on the graph



The graph crosses the x -axis at $x = 2$ and $x = 3$.



Key Takeaway

Factors $\longleftrightarrow$ **Zeros**

The factors of a polynomial give us the zeros of the function, and the zeros tell us where the graph crosses the x -axis.



Quick check

Find the zeros of the function.

$$f(x) = x^2 + 4x - 5$$

Hint: Factor the trinomial, then set each factor equal to zero.

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Graph clarification: a real zero is where the graph meets the x -axis. It can cross or touch and turn. Nonreal zeros are not x -intercepts on a real coordinate graph.

Image 7 practice: Finding zeros from factors

Q31. Find all zeros of $f(x) = x^2 - 10x + 21$.

Q32. Find all zeros of $f(x) = 2x^2 - 7x + 3$.

Q33. Find all zeros of $f(x) = x^3 - 16x$.

Q34. Find all zeros of $f(x) = x^3 - 3x^2 - 4x + 12$.

Q35. For $f(x) = (x + 5)^2(x - 2)$, list the zeros with multiplicities, the x-intercepts, and whether the graph crosses or touches at each.

3. Remainder Theorem

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What is the Remainder Theorem?

The Remainder Theorem says:

When a polynomial $f(x)$ is divided by $x - a$, the remainder is $f(a)$.

So instead of doing polynomial long division, you can simply substitute a .



Key Idea

- If a polynomial $f(x)$ is divided by $x - a$, the remainder is $f(a)$.
- You can find the remainder by evaluating the function at $x = a$.

In symbols:

$$f(x) \div (x - a) \rightarrow \text{remainder} = f(a)$$

Example 1

Find the remainder.

Suppose

$$f(x) = x^3 + 2x^2 - 5x + 3.$$

What is the remainder when dividing by $x - 2$?

Solution

Step 1: Identify a .

$$x - 2 = x - a, \text{ so } a = 2.$$

Step 2: Evaluate $f(2)$.

$$\begin{aligned} f(2) &= 2^3 + 2(2^2) - 5(2) + 3 \\ &= 8 + 8 - 10 + 3 \\ &= 9 \end{aligned}$$

Step 3: Conclusion.

The remainder is **9**.

You did not need to perform polynomial division!



Key Takeaway

The Remainder Theorem gives a fast and easy way to find the remainder when a polynomial is divided by $x - a$:

$$\text{Remainder} = f(a)$$



Quick check

Find the remainder when

$$f(x) = x^3 - 4x^2 + x + 6$$

is divided by $x - 3$.

Image 8 practice: The Remainder Theorem

Q36. Find the remainder when $P(x) = x^2 + 4x - 7$ is divided by $x - 3$.

Q37. Find the remainder when $P(x) = 2x^3 - x^2 + 5x - 4$ is divided by $x + 2$.

Q38. Find the remainder when $P(x) = x^4 - 3x^2 + 2x - 5$ is divided by $x + 1$.

Q39. Find the remainder when $P(x) = 4x^3 - 2x + 7$ is divided by $2x - 1$.

Q40. When $P(x) = x^3 + kx^2 - 4x + 6$ is divided by $x - 2$, the remainder is 10. Find k .

4. Factor Theorem

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What is the Factor Theorem?

The Factor Theorem says:

$x - a$ is a factor of $f(x)$
if and only if $f(a) = 0$.

In simple language:

If plugging a into the polynomial gives zero, then $x - a$ is a factor.



Key Ideas

- The Factor Theorem is a special case of the Remainder Theorem.
- If $f(a) = 0$, then $x - a$ is a factor of $f(x)$.
- If $f(a) \neq 0$, then $x - a$ is not a factor of $f(x)$.

In symbols:

$$x - a \text{ is a factor of } f(x) \Leftrightarrow f(a) = 0$$

Example 1

Check if $x - 2$ is a factor.

Suppose $f(x) = x^3 - 4x^2 + x + 6$.

Is $x - 2$ a factor?

Solution

Step 1: Calculate $f(2)$.

$$\begin{aligned} f(2) &= 2^3 - 4(2^2) + 2 + 6 \\ &= 8 - 16 + 2 + 6 \\ &= 0 \end{aligned}$$

Step 2: Conclusion.

Since $f(2) = 0$, $x - 2$ is a factor of $f(x)$.



Key Takeaway

To check if $x - a$ is a factor of a polynomial:

1. Substitute a into the polynomial.
2. If $f(a) = 0$, then $x - a$ is a factor.
3. If $f(a) \neq 0$, then $x - a$ is not a factor.



Quick check

Is $x + 1$ a factor of
 $f(x) = x^3 + 3x^2 - 4x - 12$?

Hint: Evaluate $f(-1)$. If the result is 0, then $x + 1$ is a factor.

Image 9 practice: The Factor Theorem

Q41. Use the Factor Theorem to decide whether $x - 3$ is a factor of $P(x) = x^2 - 8x + 15$.

Q42. Use the Factor Theorem to decide whether $x + 2$ is a factor of $P(x) = x^3 + 2x^2 - 5x + 1$.

Q43. Use the Factor Theorem to decide whether $x - 2$ is a factor of $P(x) = 2x^3 - x^2 - 8x + 4$.

Q44. Find k so that $x + 3$ is a factor of $P(x) = x^2 + kx - 18$.

Q45. Find k so that $2x - 3$ is a factor of $P(x) = 2x^3 + kx^2 - 5x + 3$.

5. The Sign Can Be Confusing

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What's the issue?

When using the Factor Theorem (or Remainder Theorem), the sign can be confusing. You must use the opposite number of the constant in the factor.

**If the factor is $x - a$,
you test $f(a)$.**

**If the factor is $x + a$,
you test $f(-a)$.**



Key Idea

The Factor Theorem says:

**$x - a$ is a factor of $f(x)$
if and only if $f(a) = 0$.**

So for a factor of the form $x + a$,
rewrite it as $x - (-a)$, and test $f(-a)$.

Factor: $x - 3 \rightarrow$ test $f(3)$

Factor: $x + 3 \rightarrow$ test $f(-3)$

Factor: $x + 4 \rightarrow$ test $f(-4)$

Examples

1 Factor: $x - 3$

Test $f(3)$.

Because $x - 3 = x - a$,
we use $a = 3$.
So test $f(3)$.

2 Factor: $x + 3$

Rewrite: $x + 3 = x - (-3)$.
Test $f(-3)$.

Because $x + 3 = x - (-3)$,
we use $a = -3$.
So test $f(-3)$.

3 Factor: $x + 4$

Rewrite: $x + 4 = x - (-4)$.
Test $f(-4)$.

Because $x + 4 = x - (-4)$,
we use $a = -4$.
So test $f(-4)$.



Key Takeaway

**For a factor $x - a$, test $f(a)$.
For a factor $x + a$, test $f(-a)$.**

Always use the opposite number
of the constant in the factor.



Quick check

What value should you substitute
to check if each is a factor?

1 $x - 5 \rightarrow$ test $f(\text{ ? })$

2 $x + 5 \rightarrow$ test $f(\text{ ? })$

3 $x + 7 \rightarrow$ test $f(\text{ ? })$

Image 10 practice: Choosing the correct substitution sign

Q46. What value should you substitute to test whether $x + 6$ is a factor of $P(x)$? Explain.

Q47. Which values should you substitute for divisors $x - 8$, $x + 9$, and $3x + 2$, respectively?

Q48. Find the remainder when $P(x) = x^3 - 2x^2 + 4x + 5$ is divided by $x + 3$.

Q49. A student tests $P(4)$ to decide whether $x + 4$ is a factor of $P(x) = x^2 + 3x - 4$.
Correct the test and decide.

Q50. Use the Factor Theorem to decide whether $3x + 2$ is a factor of $P(x) = 9x^2 + 3x - 2$.

6. Using the Factor Theorem to Completely Factor a Polynomial

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What are we doing?

We use the Factor Theorem to find one factor of the polynomial, then divide by that factor (usually using synthetic division), and continue factoring the resulting polynomial until it is completely factored.



Key Idea

- 1 If $f(a) = 0$, then $x - a$ is a factor of $f(x)$.
- 2 Divide by $x - a$ (often using synthetic division).
- 3 Factor the remaining polynomial.
- 4 Repeat if needed until completely factored.
- 5 The zeros are the values of x that make each factor equal to zero.

Example 1

Completely factor the polynomial.

1 Find a factor using the Factor Theorem.

Suppose $f(x) = x^3 - 6x^2 + 11x - 6$.

Try $x = 1$:

$$f(1) = 1 - 6 + 11 - 6 = 0$$

Therefore, $x - 1$ is a factor of $f(x)$.

2 Divide by $x - 1$ (synthetic division).

1	1	-6	11	-6
	↓			
		1	-5	6
	1	-5	6	0

The quotient is $x^2 - 5x + 6$ and the remainder is 0.

3 Factor the quadratic.

$$x^2 - 5x + 6 = (x - 2)(x - 3)$$

Numbers that multiply to 6 and add to -5 are -2 and -3 .

4 Write the complete factorization.

$$x^3 - 6x^2 + 11x - 6 = (x - 1)(x - 2)(x - 3)$$

5 Find the zeros.

Set each factor equal to zero:

$$x - 1 = 0 \quad x - 2 = 0 \quad x - 3 = 0$$

The zeros are $x = 1, 2, 3$.



Key Takeaway

Use the Factor Theorem to find one factor, then divide and factor the rest, until the polynomial is completely factored. The zeros are the values of x that make each factor zero.



Quick check

Completely factor the polynomial and find its zeros.

$$f(x) = x^3 + 2x^2 - 5x - 6$$

Hint: Try $x = 3$ first (or use the Rational Root Theorem).

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Quick-check correction: for the image's $f(x) = x^3 + 2x^2 - 5x - 6$, the hint should suggest $x = 2$, not $x = 3$. In fact, $f(3) = 24$, while $f(2) = 0$. The factorization is $(x - 2)(x + 1)(x + 3)$.

Image 11 practice: Using a verified factor to finish a polynomial

Q51. Verify that 1 is a zero of $P(x) = x^3 - 13x + 12$. Divide by the corresponding linear factor, then factor completely and find every zero.

Q52. Verify that -2 is a zero of $P(x) = x^3 - 4x^2 - 7x + 10$. Divide by the corresponding linear factor, then factor completely and find every zero.

Q53. Verify that 2 is a zero of $P(x) = x^3 + 5x^2 - 2x - 24$. Divide by the corresponding linear factor, then factor completely and find every zero.

Q54. Verify that -1 is a zero of $P(x) = x^3 - x^2 - 5x - 3$. Divide by the corresponding linear factor, then factor completely and find every zero.

Q55. Verify that $\frac{1}{2}$ is a zero of $P(x) = 2x^3 - 5x^2 - 14x + 8$. Divide by the corresponding linear factor, then factor completely and find every zero.

7. The Big Algebra 2 Connection

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The Big Picture

These ideas are all connected. For a polynomial, you can go from the expression to its factors, to its zeros, and then to the x -intercepts of its graph.

Polynomial

An algebraic expression in x (e.g., degree 2, 3, ...).



Factor

Write the polynomial as a product of simpler factors (e.g., $x - a$).



Zero

Set each factor equal to zero and solve for x .



x -intercept

Each zero gives an x -intercept on the graph: $(a, 0)$.

Example

Follow the chain for a specific polynomial.

Suppose $f(x) = (x - 2)(x + 4)$.

1 **Polynomial** $f(x) = (x - 2)(x + 4)$



2 **Factors** $x - 2$ and $x + 4$



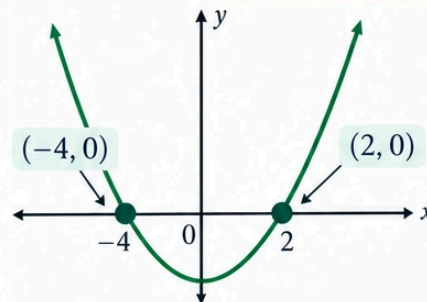
3 **Zeros**
 $x - 2 = 0 \rightarrow x = 2$
 $x + 4 = 0 \rightarrow x = -4$



4 **x -intercepts** $(2, 0)$ and $(-4, 0)$

Graph

The graph crosses the x -axis at the zeros $x = -4$ and $x = 2$.



Key Takeaway

Factoring, solving polynomial equations, the Factor Theorem, and graphs are all part of the same big idea in Algebra 2.

Polynomial $\rightarrow$ **Factor** $\rightarrow$ **Zero** $\rightarrow$ **x -intercept**

Algebra connects symbols, numbers, and graphs!



Quick check

Consider $f(x) = (x + 1)(x - 3)$.

- 1 What are the zeros?
- 2 What are the x -intercepts?
- 3 Sketch the graph (opens up or down?).
- 4 What are the factors?

Image 12 practice: Connecting expressions, factors, zeros, and graphs

Q56. For $f(x) = (x - 5)(x + 2)$, give the zeros and x-intercepts. Explain the difference between these two types of answers.

Q57. Expand $f(x) = (x - 4)(x + 6)$, then find its y-intercept and say whether the parabola opens up or down.

Q58. A quadratic has zeros -2 and 6 and passes through $(0, 24)$. Find its factored form and say whether it opens up or down.

Q59. For $f(x) = (x - 4)(x + 3)^2$, give every real zero, each x-intercept, the y-intercept, crossing or touching behavior, and end behavior.

Q60. For $f(x) = (-2x - 2)(x - 3)^2$, give every real zero, each x-intercept, the y-intercept, crossing or touching behavior, and end behavior.

Homework review: R1–R5

R1. A student factors $x^2 + 8x + 15$ as $(x + 1)(x + 15)$. Check the answer and give the correct factorization.

R2. Factor completely over the integers: $28x^3y - 42x^2y^2$.

R3. Factor completely over the integers: $3x^2 + 10x + 8$.

R4. Factor completely over the integers: $49x^2 - 100$.

R5. Factor completely over the integers: $25x^2 - 70x + 49$.

Homework review: R6–R10

R6. Factor completely over the integers: $x^3 + 6x^2 - 4x - 24$.

R7. Find all zeros of $f(x) = 3x^2 - 15x$.

R8. Find the remainder when $P(x) = 2x^3 + 3x^2 - x + 8$ is divided by $x + 2$.

R9. Find k so $x - 4$ is a factor of $P(x) = x^2 + kx - 20$.

R10. Which substitution tests the factor $4x + 3$? For $P(x) = 16x^2 - 9$, carry out the test.

Homework review: R11–R12

R11. Verify that 2 is a zero of $P(x) = x^3 - 4x^2 + x + 6$. Divide by the corresponding linear factor, then factor completely and find every zero.

R12. For $f(x) = (-x - 4)(x - 2)^2$, give every real zero, each x-intercept, the y-intercept, crossing or touching behavior, and end behavior.

Homework word problems: W1–W5

W1. A rectangular display has area $x^2 + 10x + 21$ square meters and one side $x + 3$ meters, where $x > 0$. Factor the area to find the other side. Give both dimensions when $x = 2$.

W2. A square courtyard has side x meters. A square flower bed of side 6 meters is removed from its usable area. Factor the remaining area and find it when $x = 10$, assuming $x > 6$.

W3. A square art panel has area $9x^2 + 30x + 25$ square centimeters, where $x \geq 0$. Factor the expression to find the panel's side length. What is its perimeter when $x = 3$?

W4. A fundraiser models profit by $P(n) = -n^2 + 18n - 65$ dollars when n bundles are sold, where n is a whole number from 0 to 18. Find both break-even sales levels and the whole-number sales levels with positive profit.

W5. A calibration model is $E(s) = s^3 - 5s^2 + 2s + 8$, where $0 \leq s \leq 5$. A technician uses division by $s - 2$ to test whether setting 2 has zero error. Find the remainder, then factor the model to find every allowed zero-error setting.

Homework word problems: W6–W8

W6. A rectangular storage box has positive inside dimensions x , $x + 2$, and $x + 4$ centimeters and volume 105 cubic centimeters. Form a polynomial equation, verify that $x = 3$ is a solution, and show why it is the only real solution.

W7. Over $0 \leq t \leq 6$ seconds, a moving marker's signed displacement from a reference line is $d(t) = t^3 - 7t^2 + 11t - 5$ centimeters. Verify that $t = 1$ is a zero, then find all times it meets the line. Decide whether it crosses or only touches at each time, and find $d(3)$.

W8. A designer wants a least-degree polynomial profile that crosses a reference line at $x = -2$ and $x = 5$, touches it at $x = 1$, and passes through $(0, 20)$. Negative horizontal coordinates are allowed. Find its factored equation, its y-intercept, and its end behavior.